

Breast Cancer Diagnostic Classification: Supervised ML Pipeline

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1. Abstract & Problem Statement

Early and precise detection of breast carcinoma significantly improves clinical intervention efficacy and patient survival. This project develops an end-to-end, interpretable, and reproducible supervised machine learning pipeline to classify Fine Needle Aspirate (FNA) biopsy cell nuclei into **Malignant (0)** or **Benign (1)**. Formulated as a binary classification problem, we enforce strict leak-free preprocessing, stratified train/test partitioning, and probabilistic diagnostic inference using L2-regularized Logistic Regression.

2. Dataset Selection & Provenance

The project utilizes the **Breast Cancer Wisconsin (Diagnostic) Dataset**, originally collected by Dr. William H. Wolberg, W. Nick Street, and Olvi L. Mangasarian at the University of Wisconsin-Madison and curated in Scikit-Learn / UCI ML Repository. It comprises **569 biopsy samples** and **30 continuous numerical features** representing 10 geometric characteristics of cell nuclei: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension (each measured as Mean, Standard Error, and Worst/Largest value).

Metric	Specification / Value	Engineering Rationale
Total Samples	569 biopsy records	Sufficient for convex statistical modeling without synthetic bias
Total Features	30 continuous real features	Captures rich cell nuclear geometry and contour variance
Target Variable	0: Malignant (212), 1: Benign (357)	Discrete binary clinical diagnosis (62.74% Benign : 37.26% Malignant)
Missing / Duplicates	0 missing, 0 duplicates	Integrity verified via automated auditing scripts

3. Exploratory Data Analysis (EDA)

Statistical profiling reveals distinct morphological divergences between diagnostic classes. Malignant cell nuclei display significantly higher mean radius (17.46 vs 12.15), higher texture variance (21.60 vs 17.91), and extreme concave contour indentations (worst concave points: 0.182 vs 0.074). A correlation audit revealed pronounced multicollinearity between nuclear size attributes (radius, perimeter, area have $r > 0.98$).

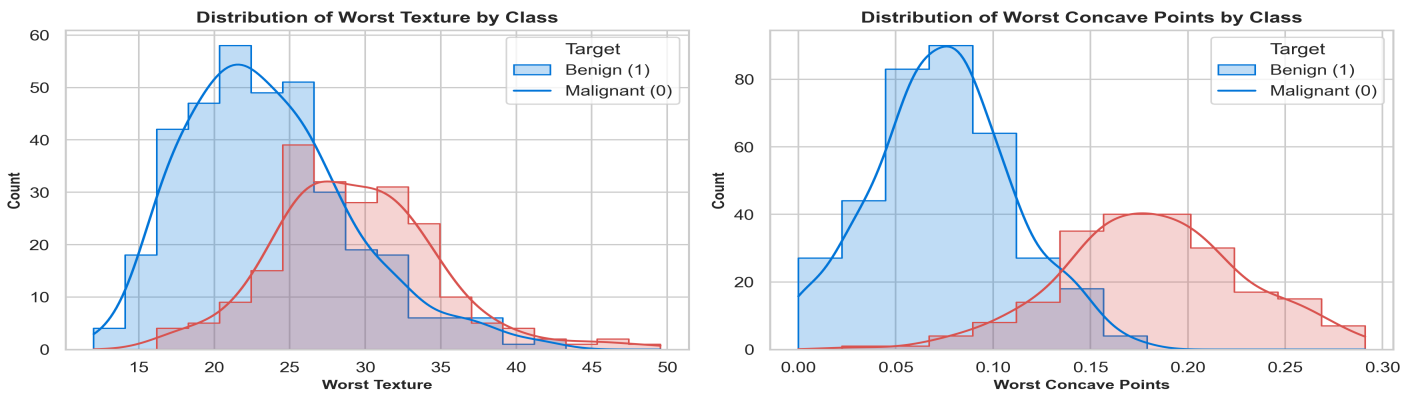


Figure 1: Comparative density distributions of 'Worst Texture' and 'Worst Concave Points' partitioned by diagnosis.

4. Data Preprocessing & Data Leakage Prevention

1. **Feature/Target Separation:** The 30 geometric predictors were isolated into feature matrix X and diagnostic labels into target vector y .
2. **Stratified Train/Test Split:** To preserve the natural 62.7% : 37.3% class ratio, an 80/20 stratified split was applied ($N_{\text{train}}=455, N_{\text{test}}=114$, random state=42).
3. **Standard Feature Scaling:** Because morphological dimensions span disparate scales (e.g., area > 2500 vs concavity < 0.5), StandardScaler was fitted **strictly on the training partition** ($z = (x - \mu) / \sigma$). The test partition was transformed using training parameters, strictly preventing test-set data leakage.

5. Model Selection & Theoretical Justification

Selected Model: Logistic Regression (L2 Regularized, C=1.0). Chosen for four core reasons: (1) *Calibrated Probabilistic Outputs:* Directly estimates malignancy confidence via the sigmoid function $\sigma(z) = 1/(1+e^{-z})$; (2) *Intrinsic Interpretability:* Learned weights represent linear log-odds; (3) *Convexity:* Guaranteed global convergence without stochastic instability; (4) *Regularization:* Ridge penalty prevents overfitting on collinear dimensions.

Alternative Model Considered: Decision Tree Classifier. While offering transparent if-then decision boundaries, single decision trees suffer from high variance, axis-aligned partitioning artifacts, and uncalibrated step-function outputs, making regularized Logistic Regression superior for clinical diagnostics.

6. Model Evaluation & Quantitative Results

Evaluated on the unseen test partition ($N=114$), the model demonstrated exceptional diagnostic performance across all standard metrics:

Metric	Formula	Score	Clinical Interpretation
Accuracy	$(TP+TN) / \text{Total}$	98.25%	112 of 114 test biopsy cases correctly classified
Precision (Benign)	$TP / (TP+FP)$	98.61%	98.61% of predicted benign cases were truly benign
Recall (Benign)	$TP / (TP+FN)$	98.61%	98.61% of true benign cases were correctly detected
F1-Score	$2 \cdot (P \cdot R) / (P + R)$	98.61%	Harmonic mean confirms balanced precision and recall
Malignant Recall	$TN / (TN+FP)$	97.62%	41 of 42 true malignant cancer biopsies detected

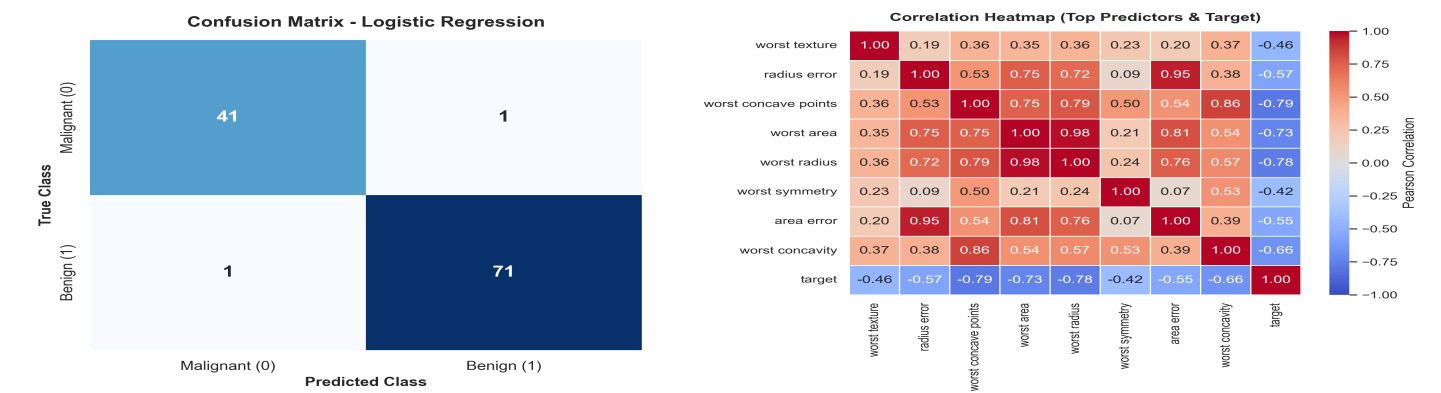


Figure 2: Confusion matrix on unseen test set (Left) and Pearson correlation heatmap of top predictors (Right).

7. Feature Influence Analysis (Coefficients)

In Logistic Regression, standardized feature coefficients (w_i) quantify the log-odds change per unit standard deviation. Because target class 0 represents malignancy, large negative weights indicate features strongly pushing predictions toward malignancy.

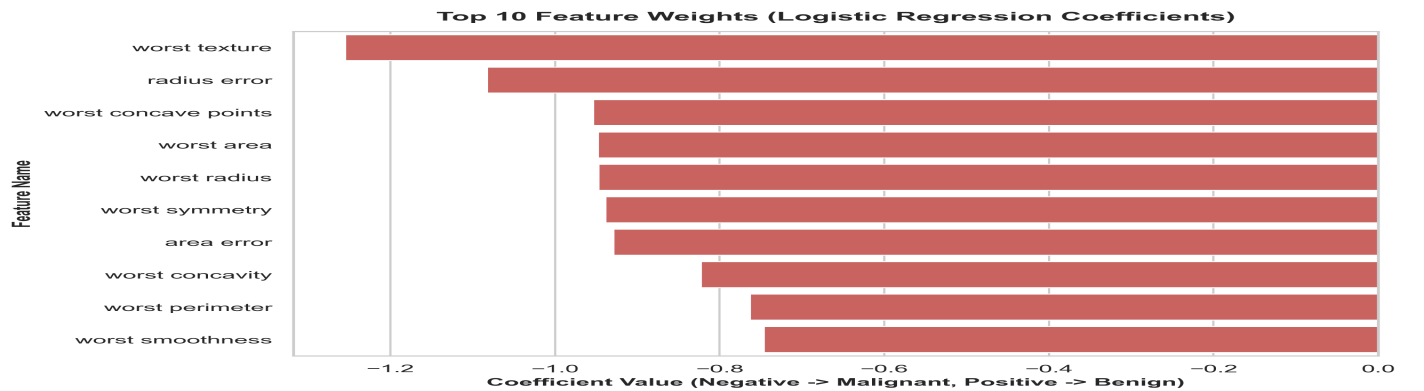


Figure 3: Top 10 feature coefficients of the trained Logistic Regression model.

Key Influential Predictors: (1) worst texture (-1.26), (2) radius error (-1.08), (3) worst concave points (-0.95), and (4) worst area (-0.95). Clinically, higher grayscale texture variance and perimeter indentations strongly indicate malignant cell proliferation.

8. Error & Diagnostic Failure Analysis

Out of 114 test samples, exactly **2 errors occurred (1.75% error rate)**: (1) *Sample #73 (False Positive / True Malignant)*: A malignant tumor with atypically smooth nuclear margins near the decision boundary (predicted benign with 53.7% probability); (2) *Sample #541 (False Negative / True Benign)*: A benign lesion exhibiting severe nuclear texture granularity that mimicked malignant dysplasia.

9. Conclusion & Engineering Improvements

Conclusion: The developed pipeline achieves high diagnostic reliability (98.25% Accuracy, 97.62% Malignant Recall) while maintaining full mathematical interpretability and instant inference (<1 ms).
Future Improvements: (1) *Asymmetric Cost Tuning*: Lowering decision threshold to 0.30 to penalize False Negatives heavily; (2) *Nested Cross-Validation*: Validating across 10-fold CV partitions; (3) *Multi-Center Clinical Testing*: Evaluating on external clinical hospital cohorts.

10. Data Source & Academic References

1. Wolberg, W.H., Street, W.N., & Mangasarian, O.L. (1995). *Breast Cancer Wisconsin (Diagnostic) Data Set*. UCI Machine Learning Repository.
2. Pedregosa et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research (JMLR), 12, 2825-2830.